

VRScroll: A Shape-Changing Device for Precise Sketching in Virtual Reality

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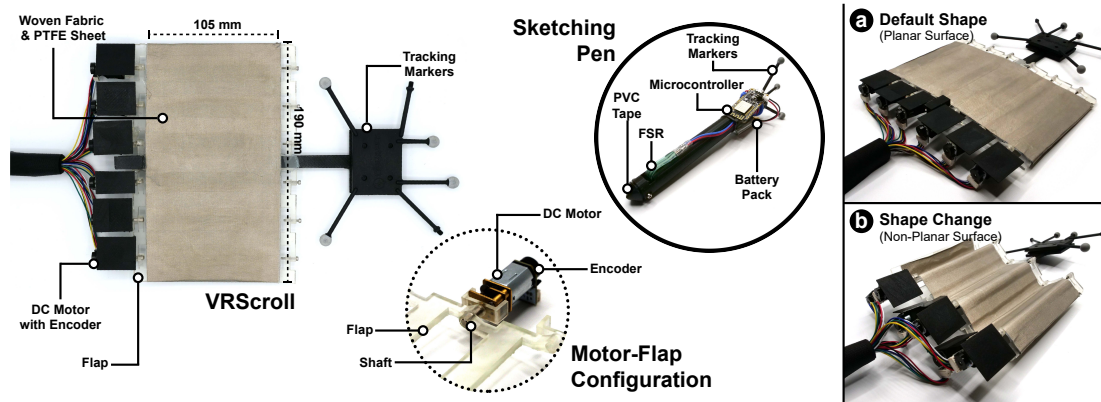


Figure 1: VRScroll device. Example surfaces created by VRScroll: (a) flat surface, (b) zigzag-fold surface.

ABSTRACT

Studies have shown that incorporating physical surfaces in VR can improve sketching performance by allowing users to feel the contact and rest their pens on it. However, using physical objects or devices to reproduce the diverse shapes of virtual model surfaces can be challenging. We propose VRScroll, a shape-changing device that physically renders the shape of a virtual surface for users to sketch on it using a pen. By providing users with passive haptic feedback and constrained movement, VRScroll has the potential to facilitate precise sketching on virtual surfaces.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices

1 INTRODUCTION

The large and immersive 3D workspace provided by Virtual Reality (VR) environments allows people to create and review 3D designs efficiently, and more VR applications, such as Tilt Brush by Google and Gravity Sketch, are developed to support such tasks. While using a pen allows users to better interact with the virtual environment, more complicated manipulations, such as precise sketching and annotating in VR, remain a major challenge as the interaction is made in the air without proper haptic feedback or movement constraints.

The importance of having a physical surface for high-precision drawing in 3D space has been demonstrated in a study by Arora et al. [2]. The study results showed that having a physical surface can help the user sketch faster and more precisely in VR. Based on this finding, researchers have developed various methods that use physical interaction surfaces, such as graphic tablets [3] or touchscreen devices [1]. However, the interaction surfaces on these devices are flat, which may lead to user frustration when the user needs to draw on a virtual object with non-planar surfaces.

Several methods have been developed to enable more precise interactions on non-planar surfaces in VR. Arora et al. developed a system utilizing indirect mappings between the virtual and physical surfaces, allowing the user to sketch on a flat physical surface and see the sketch appear in real-time on a non-planar virtual surface [1]. Elsayed et al. developed a drawing pen equipped with pneumatic and vibrotactile actuators that simulates the feeling of a pen touching a physical surface in VR [4]. In a study by Wacker et al., the researchers used 3D-printed objects that matched the virtual objects for the user to use as a physical proxy to sketch on [6]. While these methods have their own benefits, none of these provides the feeling of pen contact and constraints pen movement without the need for fabricating physical objects.

In this paper, we present VRScroll, a scroll-like hand-held shape-changing device that can change the shape of its surface to mimic the surface of a virtual object. Different from shape-changing devices developed for more natural and immersive experiences [5], VRScroll focuses on providing the physical surface to enable precise sketching on virtual objects with surfaces in various shapes. As shown in Figure 1, VRScroll has seven 3D-printed flaps controlled by DC motors to transform the device into various shapes, such as a flat surface (Figure 1a), a corner surface, or a zigzag-fold surface (Figure 1b). While VRScroll cannot render the exact shape of all virtual surfaces due to its low spatial resolution (7 flaps), we believe that the physical approximation of the virtual object surface will provide the movement constraint and the feeling of contact required for precise sketching to improve the sketching performances in VR.

2 VRSCROLL

To achieve the goal of providing a physical surface for users to make precise sketches in VR, we set the following design considerations for VRScroll: 1) the device should have an interaction surface large enough for sketching; 2) the device should be lightweight and easy to hold; 3) the device should be able to detect light contacts of a pen for smooth drawing; 4) the device should be able to render surfaces with various angles; and 5) the shape change should be quick enough to reflect changes in the virtual object in real-time.

The design of the shape-changing device was inspired by the traditional drawing surface, paper, that people fold, bend, and attach

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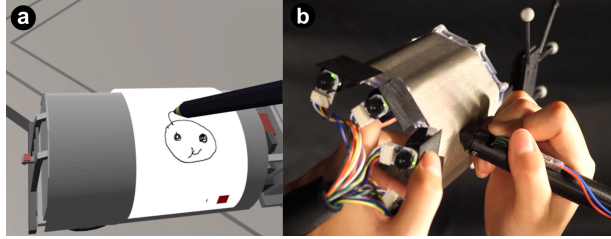


Figure 2: Sketching with VRScroll: (a) A user is sketching on the curved surface of a virtual truck in VR. (b) VRScroll changes its shape to mimic the curvature of the virtual surface.

to other surfaces. To make the drawing surface automatically change its shape based on the shape of the virtual object, we employed a mechanism that uses a set of flaps controlled by motors, similar to the mechanism used for HaptoBend [5].

To make the device lightweight and support a wide range of rotation angles between the flaps, VRScroll exploited concentric-driven links as shown in Figure 1 to connect the seven flaps fabricated with resin material using a stereolithography (SLA) 3D printer. Each link was controlled by a DC motor with an encoder (Adafruit N20, 100 rpm, 1:100 gear ratio) that supports PID control, three motor drivers (Adafruit TB6612) were used to drive the six motors, and all motors were controlled by two Arduino UNO WIFI REV 2 boards. All motor drivers and microcontrollers were powered by a 2S Li-Po battery (7.4 V, 450 mAh). The dimension of the prototype is 190 mm $\times$ 120 mm $\times$ 4 mm and the weight is 156.5 g. The device can rotate all flaps between -135 to 145 degrees, enabling the rendering of various shapes such as flat (Figure 1a), zigzag-fold (Figure 1b), and curved surfaces (Figure 2b). A 190 mm $\times$ 105 mm flexible polytetrafluoroethylene (PTFE) sheet and a woven fabric covered the flaps to create a smooth and continuous sketching surface.

To accurately detect contact between the surface and the pen in VR, we implemented a drawing pen with a force sensing resistor (FSR) attached to the tip, as illustrated in Figure 1. The pen was designed as a cylinder with a height of 170 mm and a radius of 8.5 mm, weighing 47.0 g. A black PVC tape covered the pen tip for smooth drawing. The pen tip could move vertically within the pen body. When pressure is applied to the pen tip, the pen tip is pushed up and presses the force sensor. An Adafruit Feather HUZZAH ESP8266 board powered by a 3.7 V Li-Po battery (900 mAh) was used to measure the force and send it to the PC via WiFi, where the sketch application was running.

3 SKETCHING APPLICATION

To demonstrate the use of VRScroll for sketching in VR, we developed a 3D design review application, where users can select different virtual objects and review them by sketching on their surfaces using VRScroll and the pen (see Figure 2). An OptiTrack optical motion capture system was used to track the positions and orientations of the user's VR headset, VRScroll, and the force-sensitive pen in 6-DOF.

If no object is selected, VRScroll remains as a flat surface and is shown as a flat rectangular surface in VR. When a virtual object and its surface are selected for drawing, the application estimates the angle to mimic the surface curvature and sends the sequence of angles to VRScroll. Once the sequence is sent to VRScroll, the device can change its shape within 100 ms. While the drawing surface is selected, the selected object will move with the movement of VRScroll as if the user is holding the object in hand.

The selected virtual surface gets covered with a drawing canvas with 1024 $\times$ 1024 pixels, allowing for a high-resolution rendering of the user's sketch. During sketching, ray-casting methods were used to calculate the position of the pen on the surface in VR. When the

pen touched the surface of VRScroll, a ray would be cast from the user's eye, passing through the pen tip, to the virtual surface, marking the collision position as the contact point. Then the corresponding UV coordinates of the contact point on the texture were calculated and the UV map was updated in real-time to display the sketch.

4 DISCUSSION AND FUTURE WORK

A previous study has demonstrated the advantages of utilizing a physical surface for sketching in VR [2]. However, the current methods for sketching on a non-planar virtual surface either result in a significant mismatch between the physical surface and virtual surfaces [1, 3] or require additional fabrication for physical objects [6]. In contrast, VRScroll can provide a physical surface that mimics the shape of a virtual surface using its shape-changing feature. While it has a limited number of flaps and cannot accurately replicate all shapes, it may decrease the mismatch between the virtual and physical surfaces to enhance the sketch performance and provide a more realistic sketching experience in VR. The future work may include an in-depth evaluation of VRScroll to investigate the effectiveness of a physical surface that only approximates the shape of a virtual object. Besides enabling precise sketching, the program-controlled shape-changing surface may also provide ad-hoc physical support for precise interactions. For example, two adjacent flaps may be elevated to create a linear physical guide, like a ruler, that can improve the sketch precision [6], or create a boundary around the drawing surface that constrains the pen movement.

We hope the design of VRScroll will inspire designers and researchers to exploit the physical features that shape-changing devices can provide to make VR better support productivity-oriented applications that require high-precision and high-accuracy interactions.

5 ACKNOWLEDGEMENTS

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